Paper Review: Black Holes Decohere Quantum Superpositions —D. Danielson, G. Satishchandran, and R. Wald

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1. **Bibliographical information:**
   D. Danielson,G. Satishchandran, and R. Wald, Black Holes Decohere Quantum Superpositions, Int. J. Mod. Phys. D 31 (2022) 14, 2241003, [arXiv:2205.06279].

2. **Main Claim:**
   With the existence of a black hole, the state will decohere after some time even when you recombine the superposition components adiabatically. This can happen without letting the state close to the horizon.

3. **Why is this interesting?:**
   Black hole has became a very interesting object in all fields of physics. But back to its origin, GR, it reveals a extreme gravitational environment's effects. It is well known that by adiabatically recover the state people can prevent decoherence. But this work shows under a extreme gravitational condition: black hole, people's operation will fail.

4. **Guess how the result was obtained:**
   In the abstract, the authors said they will discuss the radiation of graviton into the black hole, hence the path information is detected. This tells the wave information, the superposition of states, will not be accessible. This is the same as decoherence. Same for E&M radiation. I guess the authors will develop a mathematical process to calculate this, by describing the radiation of fields' states, deduce the difference of the partial traces of the states from the beginning to into the black hole, and then get the approximate time scale for this to happen. And they will use a simplest example: Schwarzschild Black Hole, because this can make the calculation simple but expresses their ideas successfully.

5. **My guess vs. actual paper:**
   Something I didn't guess was the paper calculated the decoherence rate by defining decoherence by inner product, and calculate E&M case first, then gravity case,while they talked about gravity first in the abstract. I also didn't guess their specific method on how to calculate the decoherence rate.

$T_D^{EM} \simeq \frac{\epsilon_0 \hbar c^6 b^6}{G^3 M^3 q^2 d^2}$
$T_D^{GR} \simeq \frac{\hbar c^{10} b^{10}}{G^6 M^5 m^2 d^4}$

6. Regime of Validity:

The paper didn't get a general calculation for every case, just in the Schwarzschild black hole, so any dynamic black hole are not accessible.

The time given is order of magnitudes, not exact numbers. The results work for $d \ll b$, $b \gg 1$ (lab far away from horizon), and $\omega \ll 1$ (low frequency field), source moving quasi-statically.

The gravity is perturbation theory.

The decoherence time is the time in principle, not people really can access. People may not be able to realize the decoherence by experiments. This brings the philosophical question that is the mathematical result the fact or the observation the fact?

7. What would you do next?:

I will calculate the case of a particle with charge and mass, for example electron, and see is the time influence of gravitational decoherence small so that we can merely take the result of E &M decoherence like the paper did. Then get a slightly general relation of the total influence and the individual influence of gravity and E&M.

A natural extension is to generalize to dynamic or charged black holes.

8. Mind Palace Entry:

This paper showed a Schwarzschild black hole can decoherence quantum states of long range interactions by absorbing soft-particles even when people recombine superposition components, and gave the predicted time for this to happen.